

Biweekly Report One

Reporting Period: 24 May 2026 – 06 June 2026

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Program: M.Tech Smart Mobility and Logistics Systems

Project: Building Foundation Models for Urban Mobility and Smart Cities

Things I learnt in this period are as follows,

1. Read chapter one, two and partially fifth of the book “Deep Learning on Graphs” by Yao Ma and Jiliang Tang. Few topics like Spectral Graph Theory, Graph signal processing, Chebyshev Polynomials and Spectral-based Graph Filters were studied in more depth.
2. Revised and solidified my understanding about Logistic regression, Word2vec Embeddings and Feedforward Neural Networks.

Things I plan to learn in next two weeks are as follows,

1. Complete reading chapter five of the book “Deep Learning on Graphs”.
2. Code a GCN from scratch.
3. Learn about LSTM, RNN, CNN and Autoencoders.
4. Revise backpropagation learning algorithm and the transformer architecture.
5. Read and understand the code of transformer and try to implement it..
6. Read any three papers out of following,
 1. GCN (ICLR'17) — Kipf & Welling
 2. TTM: Tiny Time Mixers (NeurIPS'24) — IBM
 3. STGCN (IJCAI'18)
 4. PatchTST (ICLR'23)
 5. DCRNN (ICLR'18)
 6. Graph WaveNet (IJCAI'19)
7. Download data and try to explore it.

During this two weeks period I encountered following difficulties,

1. The mathematics in the book is quite dense, making it challenging to read and fully understand some sections.